

Vol 7 Chapter 7 — Relativistic Electromagnetism

Keeper (author pass — deep math/physics revision)

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Chapter 7 — Relativistic Electromagnetism

Level 1 — one sentence

Maxwell's equations are *manifestly* Lorentz-covariant when written in 4-vector form $\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$, and this Lorentz covariance is not a postulate but inherits from BST's substrate symmetry $SO_0(4, 2) \subset SO_0(5, 2)$ — the substrate naturally contains conformal-Lorentz symmetry, so EM (the substrate's $U(1)$ gauge sector) automatically transforms covariantly.

Level 2 — graduate-physicist precision

7.1 The 4-current and 4-potential

In 4-vector notation:

- Position: $x^\mu = (ct, \vec{r})$
- 4-current: $J^\mu = (c\rho, \vec{J})$ — conservation $\partial_\mu J^\mu = 0$
- 4-potential: $A^\mu = (\phi/c, \vec{A})$
- Field tensor: $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$

Maxwell's equations (Chapter 2 Section 2.2):

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu, \quad \partial_{[\mu} F_{\nu\rho]} = 0$$

These are *manifestly* covariant (each side transforms the same way under Lorentz). Coulomb gauge fixing: $\partial_\mu A^\mu = 0$.

7.2 The field tensor under Lorentz boosts

For a boost with velocity v along $\hat{x}$:

$$E'_x = E_x, \quad E'_y = \gamma(E_y - vB_z), \quad E'_z = \gamma(E_z + vB_y)$$

$$B'_x = B_x, \quad B'_y = \gamma(B_y + (v/c^2)E_z), \quad B'_z = \gamma(B_z - (v/c^2)E_y)$$

So $\vec{E}$ and $\vec{B}$ mix under boost. There's no Lorentz-invariant separation between “electric” and “magnetic” — they're components of the unified field tensor $F^{\mu\nu}$.

Lorentz invariants: $-F^{\mu\nu}F_{\mu\nu} = 2(B^2 - E^2/c^2)$ - $F^{\mu\nu}\tilde{F}_{\mu\nu} = -4\vec{E} \cdot \vec{B}/c$

Both are unchanged by Lorentz transformations.

7.3 Substrate origin of Lorentz covariance

In BST: the substrate's $SO_0(5,2)$ contains $SO_0(4,2)$ as a subgroup. $SO_0(4,2)$ is the conformal group of 4D Minkowski space — generated by: - 6 Lorentz boosts/rotations - 4 translations - 4 special conformal transformations - 1 dilation

The full conformal group has 15 generators (rank-2 Cartan + 13 roots), matching $SO(4,2)$. The substrate-mechanism reading: Lorentz symmetry is automatic in BST because $SO(4,2) \subset SO_0(5,2)$, the substrate's natural group structure.

Lyra T2125 + SP-19b AB-2 (task #125 completed) develops $SO(4,2) \subset SO(5,2)$ as a foundational BST result for AdS/CFT-type framework.

7.4 Relativistic dynamics: charged particle in EM

Lagrangian for charged particle in EM:

$$L = -mc^2/\gamma - q\phi + q\vec{v} \cdot \vec{A}$$

(non-relativistic limit: $L = (1/2)mv^2 - q\phi + q\vec{v} \cdot \vec{A}$).

Relativistic equation of motion:

$$\frac{dp^\mu}{d\tau} = qF^{\mu\nu}u_\nu$$

with $u^\mu = dx^\mu/d\tau$ the 4-velocity. Reduces to Lorentz force law in 3-vector form.

7.5 Doppler effect and aberration

For a source emitting at frequency ω_0 moving with velocity $\vec{v}$ relative to observer:

Longitudinal Doppler: $\omega = \omega_0 \sqrt{(1 - \beta)/(1 + \beta)}$ for recession; reciprocal for approach.

Transverse Doppler (purely relativistic): even at 90° relative motion, $\omega = \omega_0/\gamma$. No classical analog.

Aberration: light from a star displaces by angle $\alpha = v/c$ for stellar aberration (used by Bradley 1729 to first measure c).

Substrate reading: Doppler is the substrate's K-type frequency shift under boost; aberration is the substrate's K-type angular shift. Both inherit from $SO(4,2)$ symmetry of the substrate $U(1)$ gauge sector.

7.6 Cosmological redshift

For light from distant galaxies, observed frequency redshifts: $z = (\lambda_{\text{obs}} - \lambda_{\text{emit}})/\lambda_{\text{emit}}$. Hubble's law: $v_{\text{recession}} = H_0 d \rightarrow z \approx H_0 d/c$ (linear at low z).

At high z : $1 + z = a_{\text{now}}/a_{\text{emit}}$ where a is the cosmological scale factor.

Substrate reading: cosmological redshift is the substrate's photon K-type wavelength stretching as space expands; substrate's cosmological scale-factor evolution is Volume 4 Chapters 1-4.

7.7 Worked example: Lorentz contraction of a charged rod

A rod of rest length L_0 with linear charge density λ_0 in its rest frame. Total charge $Q = \lambda_0 L_0$.

In a frame moving with velocity v along the rod: length is contracted to $L = L_0/\gamma$; charge is invariant (Q is Lorentz scalar); so linear density in moving frame is $\lambda' = Q/L = \gamma\lambda_0$.

The current $I = \lambda'v$ in the new frame; this current produces a magnetic field. The electric and magnetic fields in the boosted frame are exactly what Lorentz-transformation of $F^{\mu\nu}$ predicts (Section 7.2).

This is the deep meaning of “magnetism is relativistic electrostatics”: a moving observer’s “magnetic field” is just the rest-frame electric field viewed through Lorentz transformation.

7.8 K-audit anchors

- Lyra T2125 + SP-19b AB-2 (task #125 completed): $SO(4,2) \subset SO(5,2)$ substrate inheritance
- Chapter 2: Maxwell from substrate $U(1)$
- Volume 4 Chapter 3: relativistic mechanics in GR-substrate framework

Level 3 — 5th-grader accessibility

Einstein 1905: physics is the same in all uniformly-moving reference frames. Maxwell’s equations are *already* compatible with this — they predict the speed of light c is the same in all frames. The math: write everything as 4-vectors $(ct, \vec{r})$ and 4-tensors (like $F^{\mu\nu}$ combining $\vec{E}$ and $\vec{B}$). Then Maxwell’s equations look the same no matter how you boost. Cool consequence: a moving observer sees what’s purely electric in one frame as partly magnetic. “Magnetism IS relativistic electrostatics.” BST origin: the substrate’s symmetry group $SO_0(5, 2)$ contains $SO(4, 2)$, which is the 4D conformal group. So Lorentz invariance is automatic — built into the substrate’s geometry. EM (the substrate’s $U(1)$ gauge sector) inherits Lorentz covariance directly.

What comes next

Chapter 8 develops the Lagrangian formulation of EM — the variational principle.

Where to look this up

- Relativistic EM: Jackson Ch 11-12; Griffiths Ch 12
- $SO(4,2)$ conformal group: Penrose 2004, *The Road to Reality* Ch 33; Mack-Salam 1969
- BST anchors: Lyra T2125; SP-19b AB-2
- Volume 4 Chapter 3: GR-relativistic mechanics